Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Mean, median, and mode**

Find the mean of $4, 8, 6, 10, 7$.

Sum $= 35$. Mean $= \frac{35}{5} = 7$.

2 **[Foundational]** • **Target: Mean, median, and mode**

Find the median of $3, 9, 5, 1, 7$.

Sorted: $1, 3, 5, 7, 9$. Median $= 5$.

3 **[Foundational]** • **Target: Mean, median, and mode**

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

4 appears three times, more than any other value, so the mode is 4.

4 **[Foundational]** • **Target: Mean, median, and mode**

A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

Sum $= 80$, mean $= \frac{80}{7} \approx 11.43$. Sorted: $8, 10, 10, 11, 12, 14, 15$; median (4th value) $= 11$. Mode $= 10$ (appears twice).

5 **[Foundational]** • **Target: Range and quartiles**

Find the range of $5, 12, 8, 20, 3$.

Range $= 20 - 3 = 17$.

6 **[Foundational]** • **Target: Range and quartiles**

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.

Median (4th value) $= 7$. Lower half $2, 4, 5$: $Q_1 = 4$. Upper half $8, 9, 12$: $Q_3 = 9$.

7 **[Foundational]** • **Target: Range and quartiles**

Find the interquartile range (IQR) for the same data set.

IQR $= Q_3 - Q_1 = 9 - 4 = 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Variance and standard deviation

Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

Mean $= 6$. Squared deviations: $16, 4, 0, 4, 16$, sum $= 40$. Variance $= \frac{40}{5} = 8$.

9 [Intermediate] • Target: Variance and standard deviation

Find the standard deviation of the same data set, rounded to two decimal places.

$\sigma = \sqrt{8} \approx 2.83$.

10 [Intermediate] • Target: Variance and standard deviation

A data set has variance 25. State its standard deviation.

$\sigma = \sqrt{25} = 5$.

11 [Intermediate] • Target: Frequency tables

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

Sum $= 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19$. Total frequency $= 10$. Mean $= \frac{19}{10} = 1.9$.

12 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

13 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the median.

There are 10 values in order $1, 1, 1, 2, 2, 2, 2, 2, 3, 3$. Median $= \frac{2+2}{2} = 2$ (average of the 5th and 6th values).

14 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 7.01 Integration & Applications

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$P(\text{blue}) = \frac{3}{8}$.

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16 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

Two fair coins are flipped. Find $P(\text{both heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

18 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 2.01 Functions & Transformations

If $P(A) = 0.35$, find $P(\text{not } A)$.

$P(\text{not } A) = 1 - 0.35 = 0.65$.

19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

20 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

$P(A \text{ or } B) = 0.4 + 0.25 = 0.65$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Frequency tables

The bar chart shows a frequency table. Find the total number of data points.

a. 15 ✓

b. 6

c. 3

d. 10

$4 + 6 + 5 = 15$.

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22 [Foundational] • Target: Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.

a. 87 ✓

b. 22

c. 77

d. 97

$12 + 18 + 15 + 22 + 20 = 87$.

23 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.

a. 15

b. 18

c. 20

d. 17.4 ✓

$87 \div 5 = 17.4$.

24 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, find the range of books sold.

a. 12

b. 22

c. 8

d. 10 ✓

$22 - 12 = 10$.

25 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?

a. Wednesday

b. **Thursday** ✓

c. Tuesday

d. Friday

Thursday shows the tallest bar, 22 books.

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26 [Foundational] • Target: Frequency tables

A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

a. 10

b. 15 ✓

c. 3

d. 6

$4 + 6 + 5 = 15$.

27 [Foundational] • Target: Frequency tables

Using the same frequency table, find the mean.

a. 2.33

b. ≈ 2.07 ✓

c. 2.00

d. 1.87

Sum $= 1(4) + 2(6) + 3(5) = 31$; mean $= 31 \div 15 \approx 2.07$.

28 [Foundational] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{3}$

$\{1, 2\}$: $\frac{2}{6} = \frac{1}{3}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Basic probability

A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$ ✓

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

7 out of 11 total.

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30 [Intermediate] • Target: Basic probability

A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

a. $\frac{4}{13}$

b. $\frac{1}{4}$

c. $\frac{1}{13}$ ✓

d. $\frac{1}{52}$

4 queens out of 52: $\frac{1}{13}$.

31 [Intermediate] • Target: Basic probability

Two fair coins are flipped. Find $P(\text{at least one tail})$.

a. $\frac{1}{2}$

b. $\frac{3}{4}$ ✓

c. $\frac{1}{4}$

d. $\frac{1}{3}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = \frac{3}{4}$.

32 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{multiple of } 3)$.

a. $\frac{2}{5}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{1}{6}$

$\{3, 6\}$: $\frac{2}{6} = \frac{1}{3}$.

33 [Intermediate] • Target: Complementary and mutually exclusive events

A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a } 4\}$. Are A and B mutually exclusive?

a. **Yes** ✓

b. Only sometimes

c. No

d. Cannot be determined

$A = \{1, 3, 5\}$ and $B = \{4\}$ share no outcomes: yes.

34 [Intermediate] • Target: Independent events

A fair coin is flipped twice. Find $P(\text{both heads})$.

a. $\frac{1}{3}$

b. 1

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

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35 [Intermediate] • Target: Independent events

A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

a. $\frac{2}{13}$

b. $\frac{1}{26}$

c. $\frac{1}{13}$

d. $\frac{1}{169}$ ✓

$\left(\frac{4}{52}\right)^2 = \frac{1}{169}$.

36 [Intermediate] • Target: Conditional probability

In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.

a. 0.65

b. 0.195

c. ≈ 0.46 ✓

d. 0.3

$\frac{0.3}{0.65} \approx 0.46$.

37 [Intermediate] • Target: Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$ ✓

c. $\frac{5}{12}$

d. $\frac{4}{12}$

After removing one blue, 4 blue remain out of 11 total.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conditional probability • Cross-Lesson: 2.01 Functions & Transformations

Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B \mid A)$.

a. 0.4

b. 0.09

c. 0.5

d. ≈ 0.44 ✓

$\frac{0.2}{0.45} \approx 0.44$.

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39 [Challenging] • Target: Counting: permutations and combinations • Cross-Lesson: 3.01 Sequences & Series

In how many ways can 6 people be arranged in a line?

a. 720 ✓

b. 120

c. 36

d. 6

$6! = 720$.

40 [Challenging] • Target: The five-number summary • Cross-Lesson: 4.01 Trigonometry

Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

a. -15 **and** 49 ✓

b. 9 and 25

c. -15 and 40

d. -24 and 40

$9 - 1.5(16) = -15$; $25 + 1.5(16) = 49$.

41 [Challenging] • Target: The five-number summary • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

For the data set $10, 15, 20, 25, 30, 35, 40$, find the median.

a. 25 ✓

b. 30

c. 20

d. 27.5

With 7 values sorted, the median is the 4th: 25.

42 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 6.01 Limits & Derivatives

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84 ✓

b. 0.6

c. 0.16

d. 1.0

$\frac{150 + 100 - 40}{250} = \frac{210}{250} = 0.84$.

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43 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 7.01 Integration & Applications

A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15 ✓

c. 3

d. 30

$0.03 \times 500 = 15$.

44 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 8.01 Exponentials & Logarithms

In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60% ✓

b. 50%

c. 108%

d. 40%

$\frac{108}{180} = 60\%$.

45 [Challenging] • Target: Statistics & Probability • Cross-Lesson: 1.01 Algebra & Equations

Two fair coins are flipped. Find $P(\text{one head one tail})$.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{4}$

d. $\frac{1}{8}$

$P = \frac{2}{4} = \frac{1}{2}$.

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